

Please write your name neatly in the box.

Place a tick in the box which corresponds to your class

2025

HIGHER SCHOOL CERTIFICATE EXAMINATION – TRIAL PRACTICE II

Physics

General Instructions

- Reading Time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper

	Monday (BHHS)
	Monday (Mixed)
	Tuesday (Parramatta)
	Tuesday (Penrith)
	Wednesday (BHHS - Early)
	Wednesday (BHHS – Late)
	Wednesday (Mixed – Aaron)
	Wednesday (Mixed – Chris)
	Thursday (Ponds)
	Friday (Girraween)
	Friday (Mixed)
	Sunday (Mixed)
	Sunday (Normanhurst)

Total marks:
100

Section I – 20 marks

- Attempt Questions 1–20
- Allow about 30 minutes for this section

Section II – 80 marks

- Attempt Questions 21–34
- Allow about 2 hours 30 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

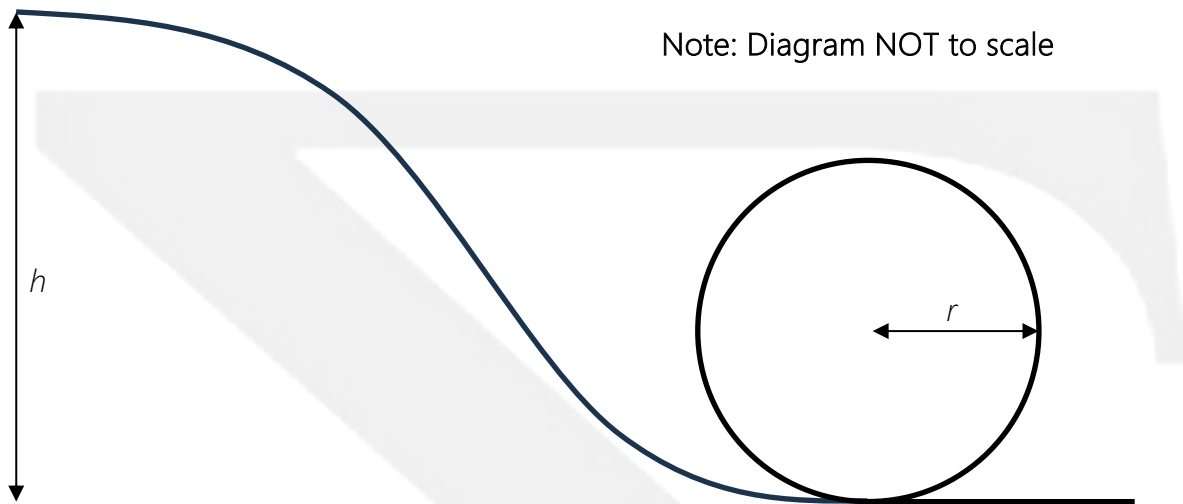
1	☐	☐	☐	☐	11	☐	☐	☐	☐
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6	☐	☐	☐	☐	16	☐	☐	☐	☐
7	☐	☐	☐	☐	17	☐	☐	☐	☐
8	☐	☐	☐	☐	18	☐	☐	☐	☐
9	☐	☐	☐	☐	19	☐	☐	☐	☐
10	☐	☐	☐	☐	20	☐	☐	☐	☐

Section I

Part A – 20 marks

Allow about 30 minutes for this part.

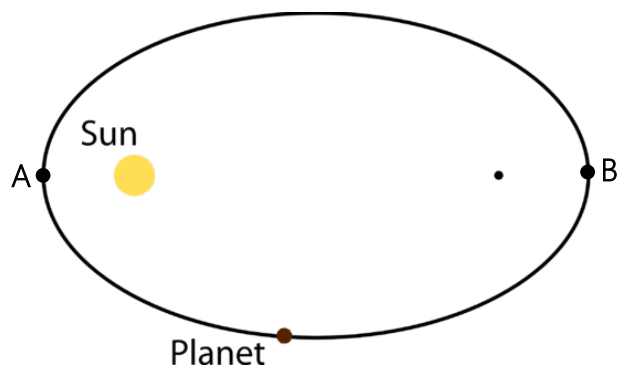
- 1 A marble is released, from rest, from a height, h , above the ground. This marble then safely completes a loop-de-loop as shown in the diagram below, with the minimum possible energy.



What is value of h , in terms of r ?

- (A) $h = \frac{10}{4} r$
(B) $h = 5r$
(C) $h = \sqrt{gr}$
(D) $h = \sqrt{5gr}$
- 2 Which of the following statements is **most correct** for a car undergoing horizontal circular motion?
- (A) Centripetal force balances frictional force
(B) Frictional force opposes weight force
(C) Centripetal force is proportional to weight force
(D) Normal force is provided by centripetal force

- 3 The diagram below shows a planet travelling around the Sun.



Which of the following best compares the linear velocity and the angular velocity of the planet at the points labelled A and B?

	<i>Linear velocity</i>	<i>Angular velocity</i>
(A)	$A > B$	$A > B$
(B)	$A > B$	$A < B$
(C)	$A < B$	$A > B$
(D)	$A < B$	$A < B$

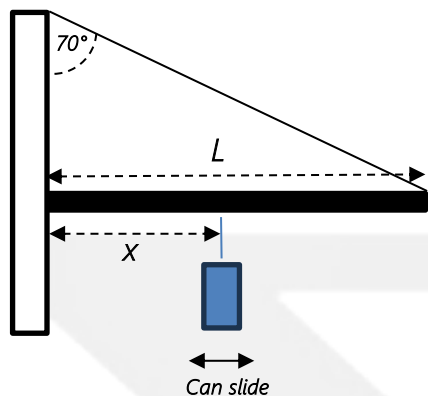
- 4 The table below lists some properties of the planet, Jupiter.

Mass	$1.898 \times 10^{27} \text{ kg}$
Surface gravity	24.79 m s^{-2}
Rotational period	9 hours 56 minutes
Orbital period	11.86 years

A **Jovistationary orbit** is a satellite orbit where the satellite would appear to remain in a 'fixed' position according to an observer on Jupiter (similar to a Geostationary satellite on Earth). The altitude of the jovistationary orbit is closest to:

- (A) $7.15 \times 10^7 \text{ m}$
- (B) $8.86 \times 10^7 \text{ m}$
- (C) $1.60 \times 10^8 \text{ m}$
- (D) $2.49 \times 10^8 \text{ m}$

- 5 A beam of mass, m , and length, L , is attached to a string which is connected to the top of a post as shown in the diagram below.
- Another hanging mass, also of mass, m , is attached to the beam and can slide horizontally along the beam.



If the beam is kept in place by the friction between the beam and the post, where co-efficient of friction is 0.5, the minimum value of ' x ' required to keep the beam locked in position is closest to?

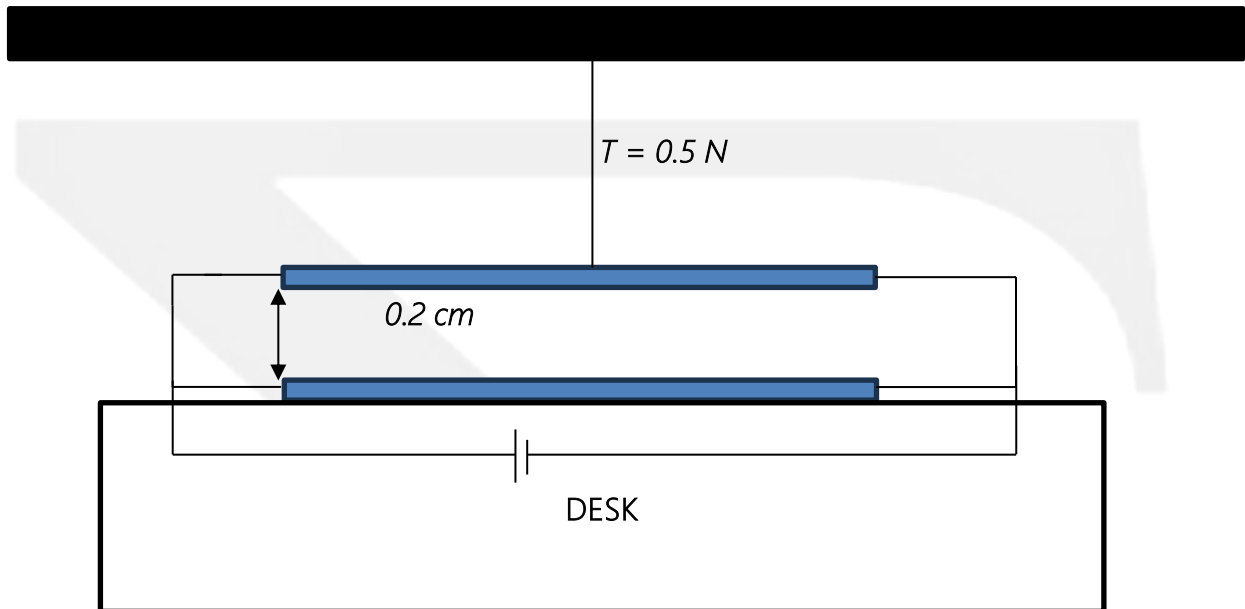
- (A) $\frac{L}{10}$
- (B) $\frac{L}{5}$
- (C) $\frac{L}{3}$
- (D) $\frac{L}{2}$
- 6 In which scenario below would the force on an electron **NOT** be zero?
- (A) When it is stationary in a magnetic field
- (B) When it is moving parallel to a magnetic field
- (C) When it is moving anti-parallel to a magnetic field
- (D) When it is travelling through perpendicular magnetic and electric fields providing forces in opposite directions at a speed equal to the ratio of the magnetic field to the electric field

- 7 Two 3.0 g wires of length 1.2 m are arranged such that they are vertically above and parallel to each other.

The bottom wire rests on the top of a desk and the top wire is connected by a string to the ceiling and they are separated by a distance of 0.2 cm.

The wires are then connected in parallel using a DC power supply as shown in the diagram below.

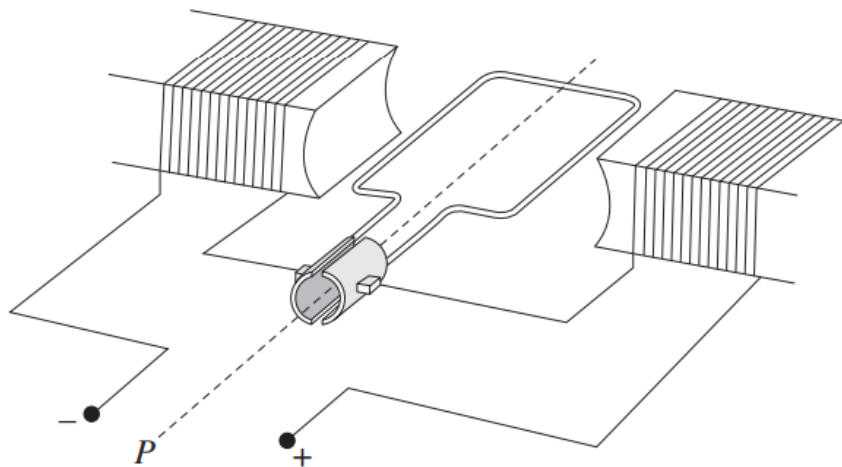
If the tension in the string is measured as 0.5 N, what must the current through the wires be?



- (A) 63 A
- (B) 65 A
- (C) 491 A
- (D) 626 A

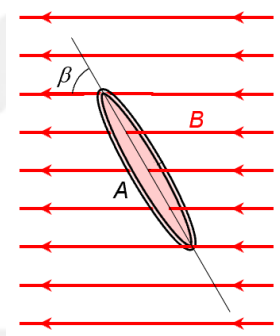
8 In the motor shown, the rotor spins clockwise as viewed from point P.

What would happen if the polarity of the DC supply is reversed?



- (A) The rotor would not spin.
- (B) The rotor would spin anticlockwise only.
- (C) The rotor would spin clockwise only.
- (D) The rotor would spin both clockwise and anticlockwise (oscillate).

9 A circular coil with n turns is placed in a magnetic field as shown below



The best expression for the total flux through the coil would be

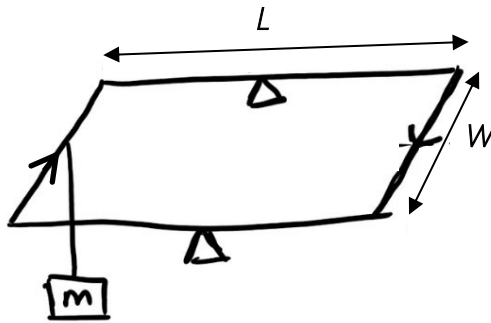
- (A) $BA\sin(\beta)$
- (B) $BA\cos(\beta)$
- (C) $nBA\sin(\beta)$
- (D) $nBA\cos(\beta)$

10 A coil carries a current, I and is placed in a magnetic field of strength, B .

The dimensions of the coil are as follows:

- length, L ,
- width, W ,

The coil is pivoted in the middle and is balanced by a mass, m , hanging on one side as shown in the diagram below.



The dimensions of the coil are doubled whilst still being pivoted in the middle and the original mass is still applied on the very left of the coil.

For the coil to remain balanced, what additional mass must be attached to the very left of the coil?

- (A) No extra mass is required.. Chris.. You troll. 🐼
- (B) m
- (C) $2m$
- (D) $3m$

11 Unpolarised light of intensity I_0 passes through 3 successive polaroids, P_1 , P_2 and P_3 . Each polaroid creates an angle of 45° to the previous. What is the intensity of the light emerging from the final polaroid?

- (A) $\frac{\sqrt{2}}{4} I_0$
- (B) $\frac{I_0}{4}$
- (C) $\frac{\sqrt{2}}{8} I_0$
- (D) $\frac{I_0}{8}$

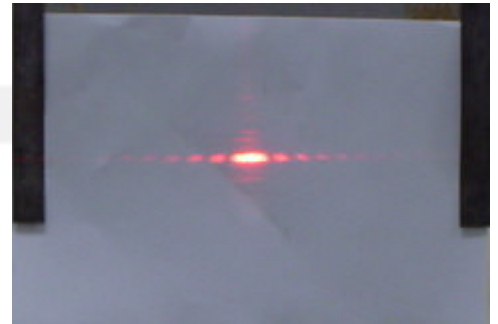
12 What was the name of the scientist who studied eclipses of Jupiter's moon(s) to determine the speed of light?

- (A) Bradley
- (B) Froome
- (C) Io
- (D) Romer

13 Light is shone through a small single slit and the number of bright spots on a screen is measured.

If the slit width is halved, which of the following would happen?

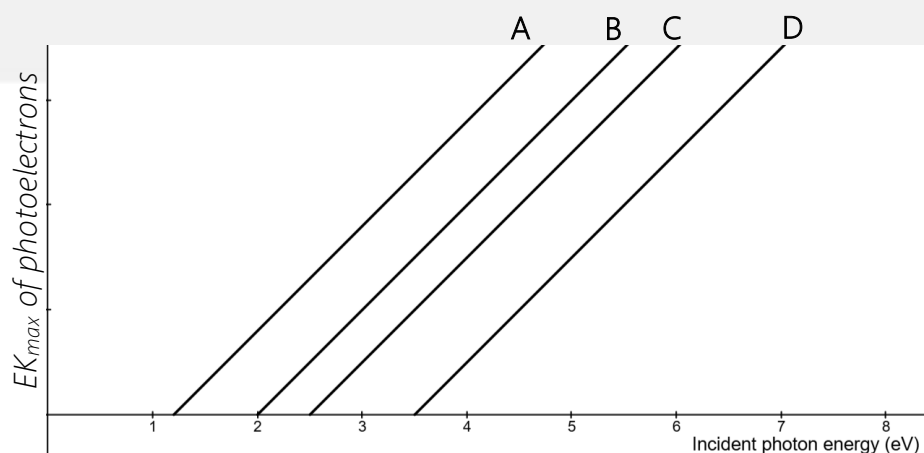
- (A) Less bright spots would appear on the screen.
- (B) More bright spots would appear on the screen.
- (C) Same number of bright spots would appear on the screen, but reduced intensity
- (D) Same number of bright spots would appear on the screen, but increased intensity.



14 In a photoelectric effect experiment, an EM wave is incident on a metal surface and electrons are ejected. It was found that the stopping voltage for the electrons is 2.5 V. If the work function of the metal is 1.64 eV, the wavelength of the incident EMR is closest to:

- (A) 200 nm
- (B) 300 nm
- (C) 400 nm
- (D) 500 nm

15 The graph below shows the relationship between maximum kinetic energy of emitted electrons and the incident energy for four different metal surfaces.



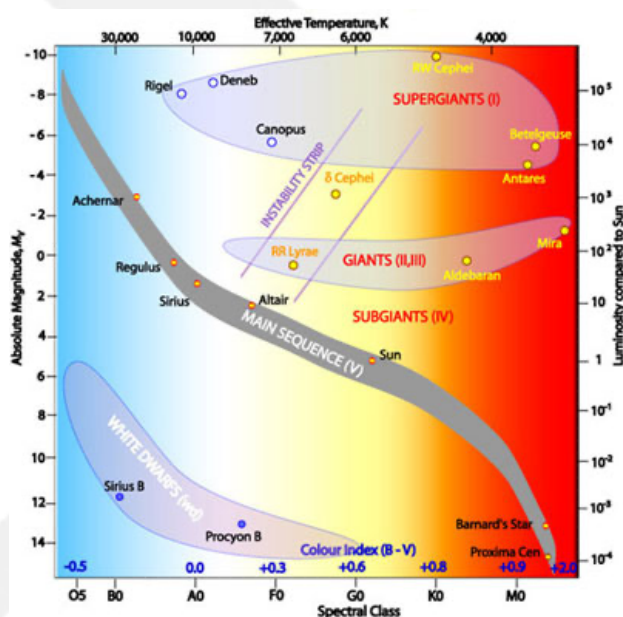
Question 15 continues on the next page

If light of wavelength 420 nm was shone onto the metals, which of the following options best represents which metals would release photoelectrons?

Note: ✓ means photoelectrons ARE released

	Metal A	Metal B	Metal C	Metal D
(A)	✓	✓	✓	X
(B)	✓	✓	X	X
(C)	✓	X	X	X
(D)	X	X	X	✓

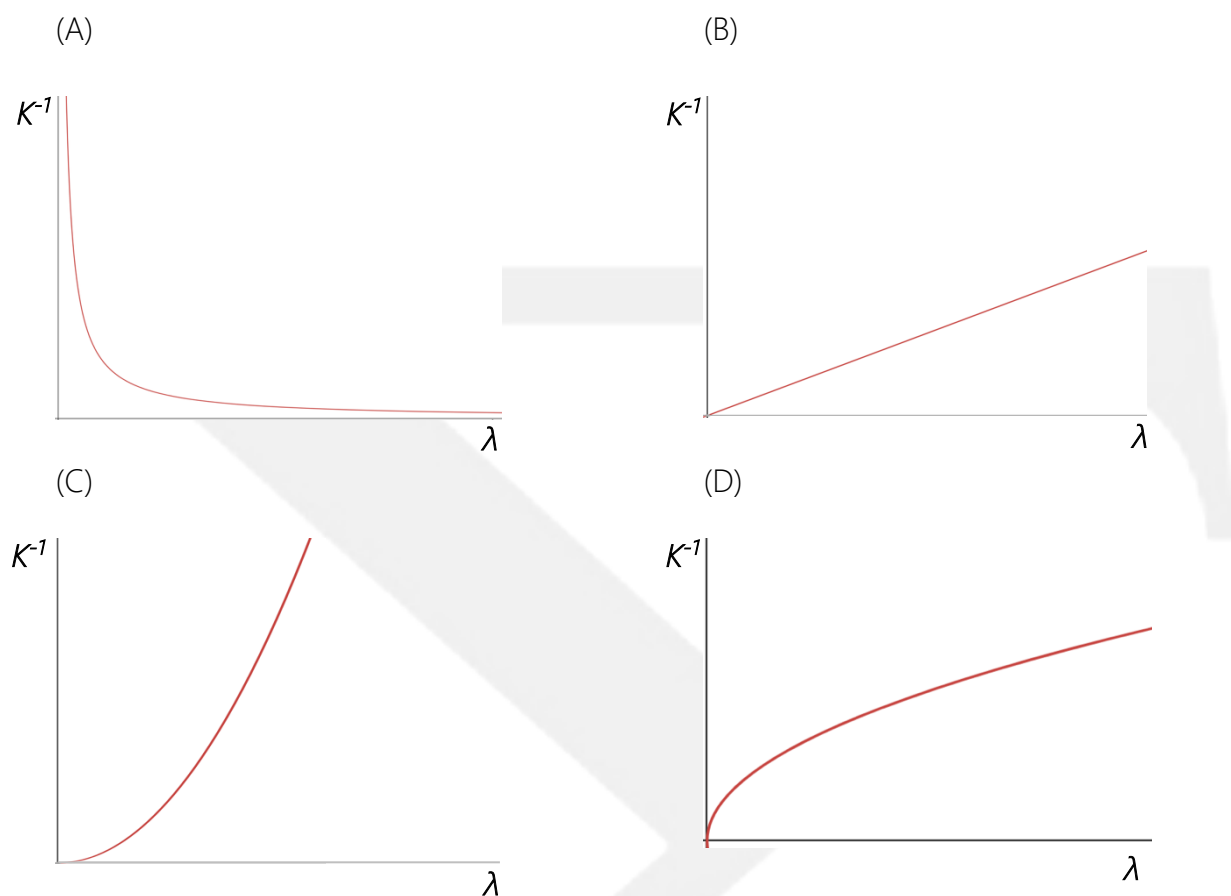
16 Using the Hertzsprung-Russell Diagram below answer the following:



Which of the options below best compares the features of the spectra of Achernar compared to Barnard's Star?

	Number of spectral lines	Peak frequency
(A)	Less lines present	Higher peak frequency
(B)	Less lines present	Lower peak frequency
(C)	More lines present	Higher peak frequency
(D)	More lines present	Lower peak frequency

- 17 If λ and K are de Broglie wavelength and kinetic energy, respectively. Which of the following graphs best shows a valid relationship between the variables for a particle of constant mass?



- 18 Which of the following was used as evidence for the existence of neutrinos?

- (A) Deep inelastic scattering
- (B) Geiger-Marsden experiment
- (C) Inverse beta decay
- (D) Rossi-Hall experiment

- 19 What is the role of the moderator in a nuclear reactor?

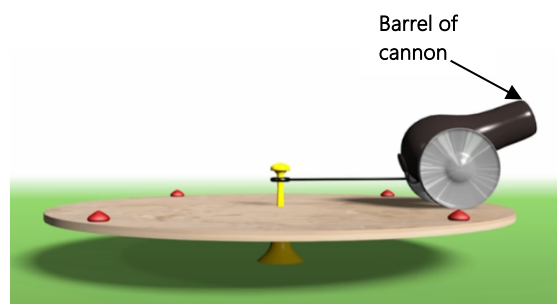
- (A) Absorb neutrons
- (B) Slow down neutrons
- (C) Speed up neutrons
- (D) Produce or release neutrons

20

A cannon is placed on the edge of a 4.5 m diameter merry-go-round which has a period of 2.0 s.

The cannon fires a cannon ball out of the barrel of the cannon at a velocity of 15 m s^{-1} @ 27° above the horizontal from a height of 25.0 m above the ground.

When the cannon ball strikes the ground, the distance between the position of the cannon ball and the position of the barrel of the cannon is closest to?



- (A) 41 m
- (B) 48 m
- (C) 53 m
- (D) 56 m

Section II

Part B – 80 marks

Allow about 2 hours 30 minutes for this part.

Question 21 (5 marks)

A rocket of mass 1200 kg is launched on an incline as shown in the diagram below.



Based on the following:

- The rocket produces thrust of 36 kN which acts for 7.0 s
- Mass of rocket remains constant
- Direction of the rocket remains constant for those 7.0 seconds
- g remains constant

Find the range of the rocket if it returns to the ground.

Question 22 (7 marks)

A car of mass 1520 kg travels around a circular track of radius 192 m with a banking angle of 7° .

(a) Derive the equation for the ideal speed of the car.

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(b) Hence, or otherwise, find the ideal speed of the car.

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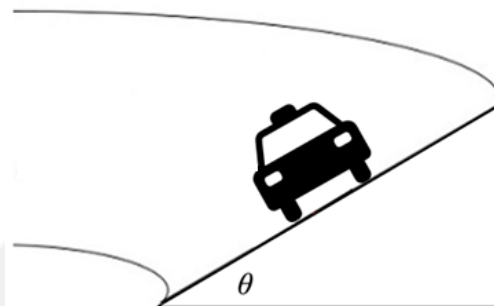
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It was found that when friction was introduced, the minimum speed the car can move around the banked track was 10 m s^{-1} slower than the speed found in (b).

(c) Sketch the force(s) acting on the car at its minimum speed in the diagram below.

2



(d) Find the coefficient of friction in the scenario above

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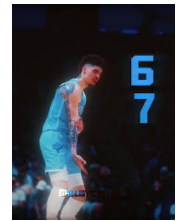
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Question 23 (7 marks)

A rocket of mass 1500 kg is launched from a position on Earth's equator to a Low Earth orbit of altitude 67 km (aka **Lamelow** Earth orbit).



- (a) Determine the change in gravitational potential energy of the rocket as it moves from Earth's surface to the Low Earth orbit. 2

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- (b) Calculate the minimum work required to safely get the rocket from Earth's surface into the low-Earth orbit. 3

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- (c) Account for any differences in (a) and (b) 2

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Question 24 (8 marks)

This question is about a method to investigate how the force on a conductor varies with flux density and current.

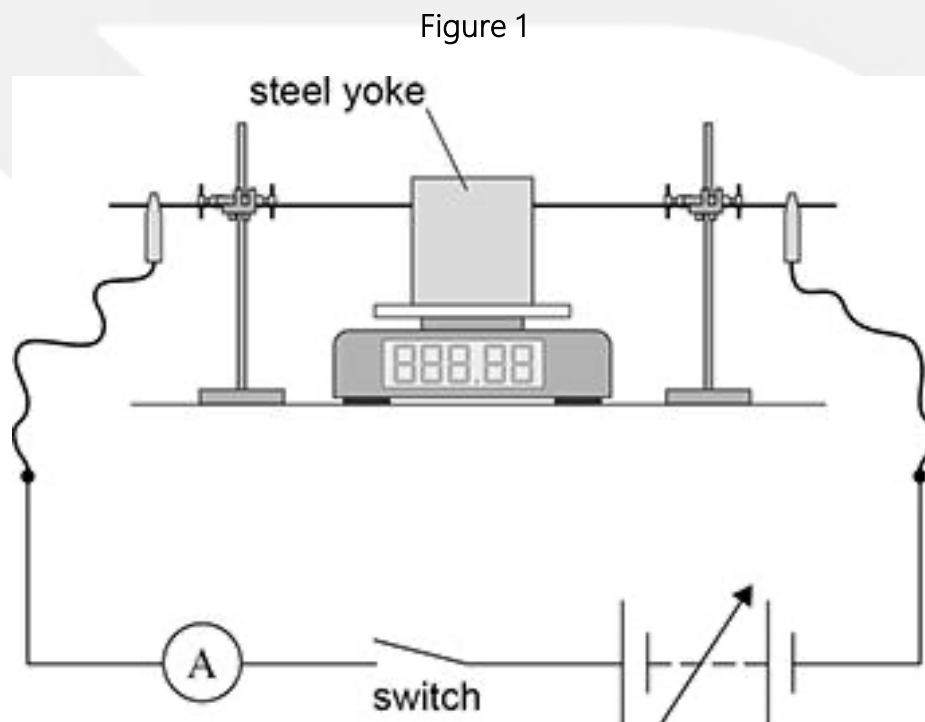
Figure 1 shows the copper rod positioned above a digital balance.

Two identical magnets are mounted on a steel yoke with their opposite poles facing each other.

The balance is zeroed.

The yoke is then placed on the balance so that a horizontal uniform magnetic field is applied perpendicular to the copper rod.

The ends of the rod are connected as shown.



- (a) When the switch is open, the reading on the balance shows the mass of the yoke and the two magnets. When the switch is closed, the reading on the balance decreases.

Identify the direction of the horizontal magnetic field.

1

Question 24 (continued...)

The current I in the rod is varied.

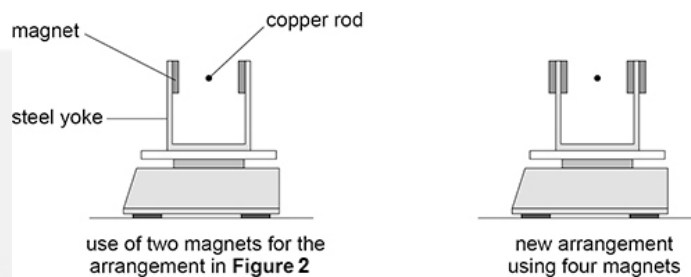
The balance reading M_1 is recorded for different values of I .

The switch is now opened.

Two additional magnets, identical to those used before, are attached to the yoke.

Figure 2 shows how this new arrangement compares with the arrangement in Figure 1.

Figure 2



The balance reading with four magnets attached to the yoke is M_2 .

With the switch open, M_2 is the mass of the yoke and the four magnets.

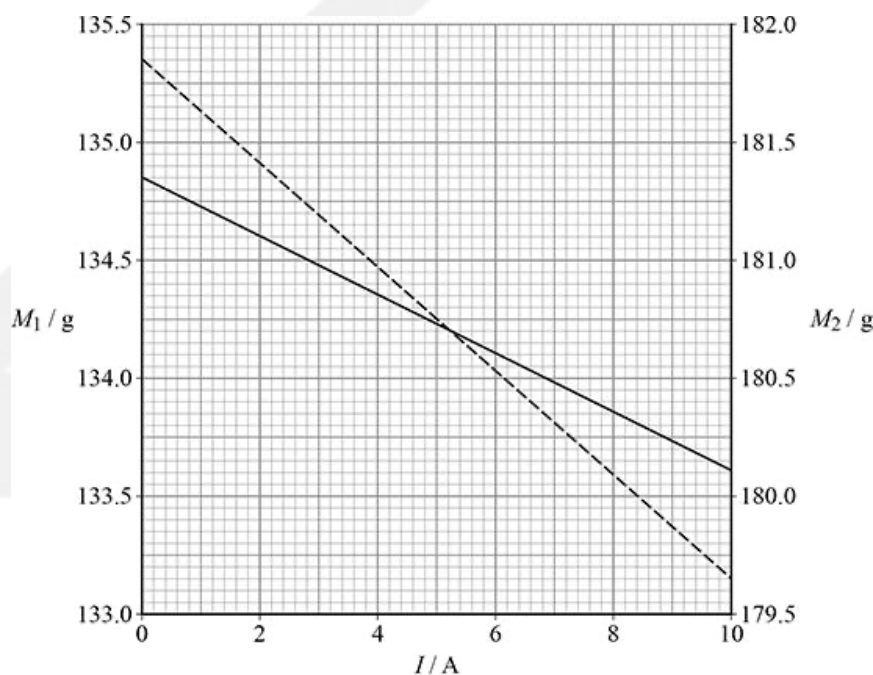
The switch is now closed.

M_2 is recorded for different values of I .

Figure 3 shows data from both experiments.

Values of M_1 and M_2 are plotted using different vertical axes.

Figure 3



The solid line — shows the variation of M_1 with I

The dashed line - - - shows the variation of M_2 with I

Question 24 (continued...)

It can be shown that

$$M = kBI + nZ + Y$$

Where:

M = balance reading when the current is I

B = magnetic flux density of the horizontal uniform magnetic field

n = number of magnets attached to the yoke

Z = mass, in g, of each magnet

Y = mass, in g, of the yoke

k is a constant.

- (b) Deduce the fundamental base units for k . 2

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- (c) Determine Y . 2

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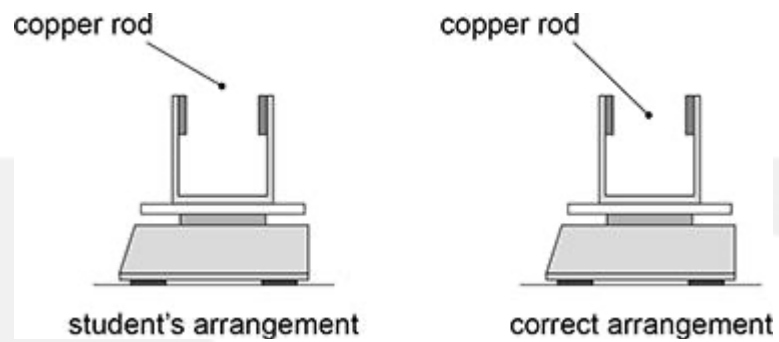
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Question 24 (continued...)

- (d) A student sets up the apparatus with the copper rod positioned incorrectly.

Figure 4 shows how the student's arrangement compares with the correct arrangement.

Figure 4



The student produces a graph of M_1 against l .

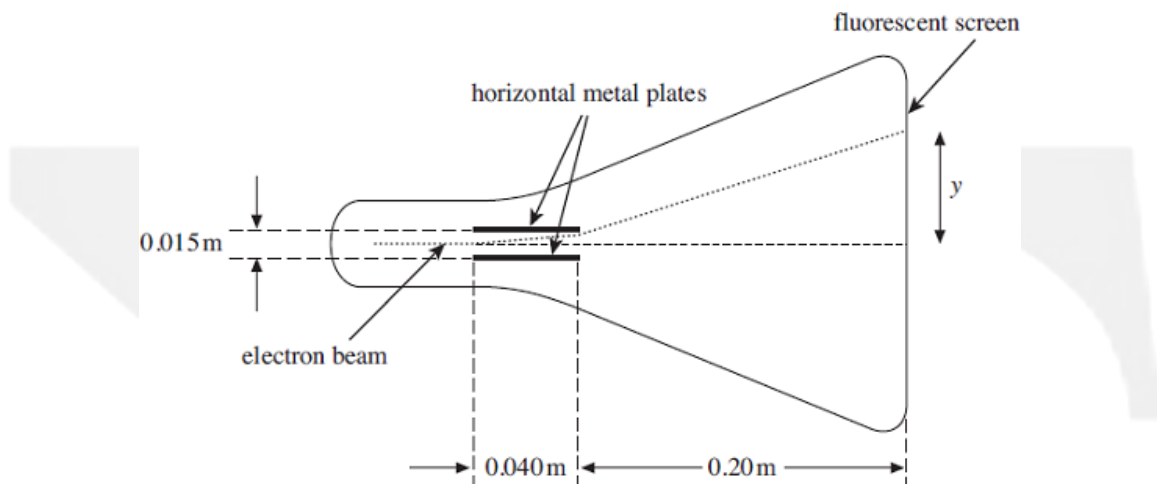
Compare the student's graph with the graph of M_1 against I (the solid line) in **Figure 4**. Explain your answer.

Question 25 (6 marks)

The figure below shows the cathode ray tube of an oscilloscope.

A potential difference of 270 V is applied between the horizontal metal plates.

An individual electron takes 1.5×10^{-9} s to pass between the plates.



- (a) Find the acceleration of the electrons as they pass through the horizontal metal plates. 2

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- (b) Find the vertical displacement, y , of the electron beam from the centre of the screen. 4

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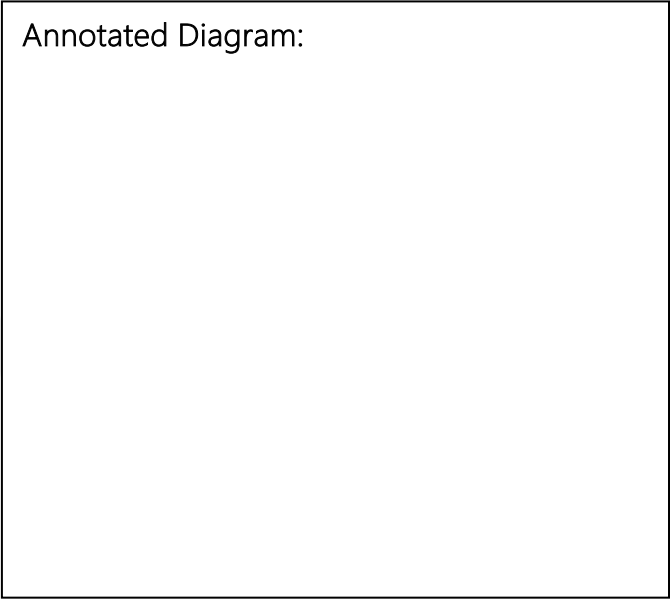
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Question 26 (7 marks)

Explain, in detail, the operation of a simple DC motor.
Include an annotated diagram of a simple DC motor to support your response.

Annotated Diagram:



Question 27 (5 marks)

Explain how rotational velocity of a star can impact its spectral lines.

Question 28 (4 marks)

A spacecraft is moving away from Earth with uniform velocity. A radio wave is sent from Earth at 3:00pm and strikes the rear of the spacecraft. The reflected signal is received on Earth at 3:02pm. The pulse which is reflected from the front of the craft is received $2\mu\text{s}$ later. A second radio wave is sent at 3:04pm and the first reflection is received at 3:18pm.

Find the speed and the proper length of the spacecraft.

Question 29 (6 marks)

(a) Outline how Newtons model of light explained reflection.

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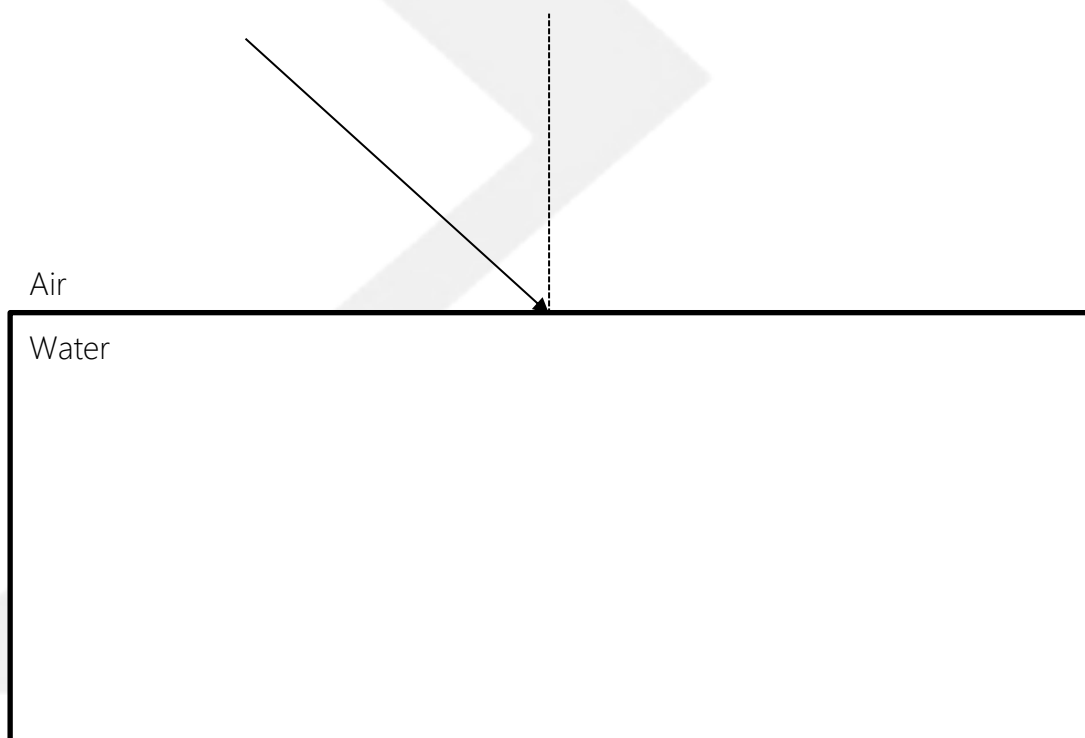
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(b) Complete a labelled diagram to show the propagation of waves as they move from air to water, according to Huygens model of light.

3



Question 30 (6 marks)

When monochromatic light is shone on a clean metal surface, electrons are emitted from the surface due to the photoelectric effect.

Explain the effect on the emitted electrons of:

(a) Increasing the frequency of light.

3

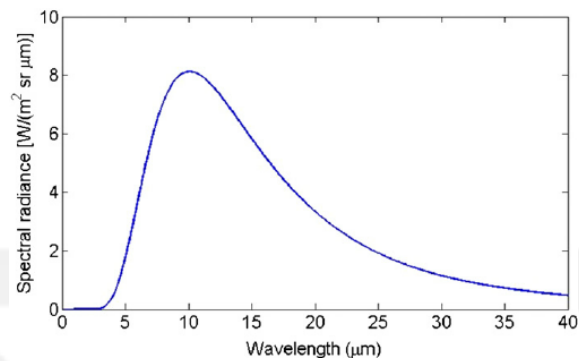
(b) Increasing the intensity of light

3

Question 31 (6 marks)

In the middle of the 20th century, there were two competing theories of the Universe. In 1964, electromagnetic radiation was observed coming from all directions in space.

The diagram below shows the distribution of this radiation as observed from Earth.



The graph provides evidence for one of these theories of the Universe.

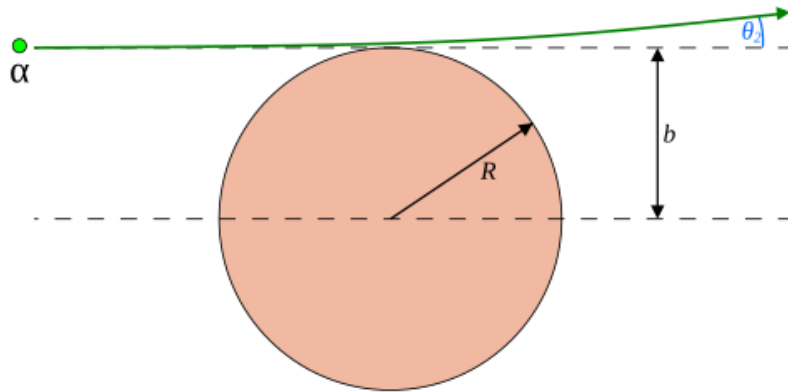
Discuss the main features of this theory of the Universe.

In your answer, you should include:

- the main predictions and evidence for the theory, and
- a suitable calculation.

Question 32 (5 marks)

An alpha particle is fired tangentially at $1.53 \times 10^7 \text{ m s}^{-1}$ towards an Au-197 nucleus. The atom can be seen as a solid, positive sphere and the impact of electrons can be neglected cuz they lowk be beta particles.



If the radius of the gold atom is 1.44×10^{-10} m, find the **approximate** angular deflection (θ_2) of the alpha particle above. List **ANY** assumptions you have made.

Question 32 (8 marks)

By referring to specific examples you have studied during HSC Physics, analyse the significance of conservation laws in developing our understanding of physics.



QUESTION
NUMBER

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QUESTION
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QUESTION
NUMBER

EXTRA WRITING SPACE
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Walkthrough video available:

<https://youtu.be/4P32t8gNHCo>